

AMAZON SALES AND OPERATIONS ANALYSIS

SQL, Python & Power BI Project Report

Business Requirements, Database and Data Model, Business Question
Insights and Interactive Dashboard Analysis

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1. Business Requirements

The business operates an Amazon-style online marketplace where customers place orders for products sold by many different sellers. Orders are paid for, shipped through delivery providers and fulfilled from several warehouses. The business wants a data analyst to study this data, answer key business questions and present the results in an interactive dashboard so that decisions can be made on sales, customers, sellers, shipping and inventory.

1.1 Business Questions (SQL Analysis)

The business requires answers to the following 20 questions, to be solved with SQL queries on the database:

- Q1.** Top 10 selling products based on orders that were not cancelled or returned (category, product name, total orders, quantity sold and sales value).
- Q2.** Revenue generated by each product category.
- Q3.** Average Order Value (AOV) for customers having at least 15 orders.
- Q4.** Month-by-month sales trend starting from the same date last year up to the latest, showing sales for the current and previous months.
- Q5.** Customers who are registered but have not placed any order yet.
- Q6.** Top 3 best-selling product categories in each state, including their sales value.
- Q7.** Least-selling product category in each state, including its sales value.
- Q8.** Total value of orders placed by each customer over their lifetime, ranked by order value.
- Q9.** Products with stock levels below a threshold (for example 10 units) to give a low stock alert.
- Q10.** Orders where the shipping delay is more than 5 days after the order was placed, with customer, order and shipping provider details.
- Q11.** Percentage of each payment status across all orders.
- Q12.** Top 5 sellers by sales with their successful orders, failed orders and success rate, plus the top 5 products of the top seller.

- Q13.** Gross profit and profit margin for each product, ranked by profit margin.
- Q14.** Top 10 most returned products and the percentage of their ordered units that was returned.
- Q15.** Sellers who have made no sales in the last 6 months, with their most recent sale date and total sales.
- Q16.** Return activity of customers who placed at least 10 orders in the last 2 years, classified as High or Normal Return Frequency.
- Q17.** Top 5 customers in each state by number of orders, with their total orders and total sales.
- Q18.** Running total of sales for each year along with the sales generated in each year.
- Q19.** Performance of each shipping provider: total revenue, orders handled and average shipping time.
- Q20.** Year-over-year revenue growth percentage for each recorded year.

In addition, the business requires an automated inventory update mechanism (a stored procedure) that decreases the stock quantity of the corresponding product whenever a new sales record is added.

1.2 Power BI Dashboard Requirements

The business also requires an interactive Power BI dashboard that turns the results of the analysis into a visual report. The dashboard must meet the following requirements:

- **Four report pages:** Executive Sales Overview, Customers & Sellers Insights, Shipping & Delivery Performance and Inventory Health Analysis.
- **Filters:** Category, Year and Month slicers so that the pages can be explored for any selection.
- **Sales KPIs:** total revenue, current revenue, year-over-year and month-over-month revenue growth, total orders, average order value and profit margin.
- **Sales views:** revenue trend over time, revenue by category, top 10 products by revenue and state-wise revenue distribution.

- **Customer and seller KPIs:** total customers, returning customers, active sellers, average revenue per seller and average revenue per customer, with views of customers by state, top customers, customer return frequency, seller origin and seller performance.
- **Shipping KPIs:** delivery, in-transit, cancel and return rates, average days in shipping, order and payment status breakdown, shipping provider performance, returns by category and the monthly return trend.
- **Inventory KPIs:** total stock in hand, products below the stock threshold, total inventory value and total warehouses, with views of stock and orders by warehouse, product stock status and stock versus sales velocity.

2. Database and Data Model

The project database contains 9 tables. They are created in two groups: 3 parent tables, which do not depend on any other table, and 6 child tables, which depend on the parent tables through foreign keys. Every table has a primary key.

2.1 Tables

Table	Columns
category	category_id (PK), category_name
customers	customer_id (PK), first_name, last_name, state, address
sellers	seller_id (PK), seller_name, origin
products	product_id (PK), product_name, price, cogs, category_id (FK)
orders	order_id (PK), order_date, customer_id (FK), seller_id (FK), order_status
order_items	order_item_id (PK), order_id (FK), product_id (FK), quantity, price_per_unit
payments	payment_id (PK), order_id (FK), payment_date, payment_status
shipping	shipping_id (PK), order_id (FK), shipping_date, return_date, shipping_provider, delivery_status
inventory	inventory_id (PK), product_id (FK), stock, warehouse_id, last_stock_date

2.2 Parent Tables

- **category:** stores the product categories.
- **customers:** stores customer details and the state of each customer.
- **sellers:** stores the sellers and their country of origin.

2.3 Child Tables

- **products:** stores product details with selling price and cost of goods sold (cogs), and links each product to a category.
- **orders:** stores each order with its date and status, and links it to the customer who placed it and the seller who fulfilled it.

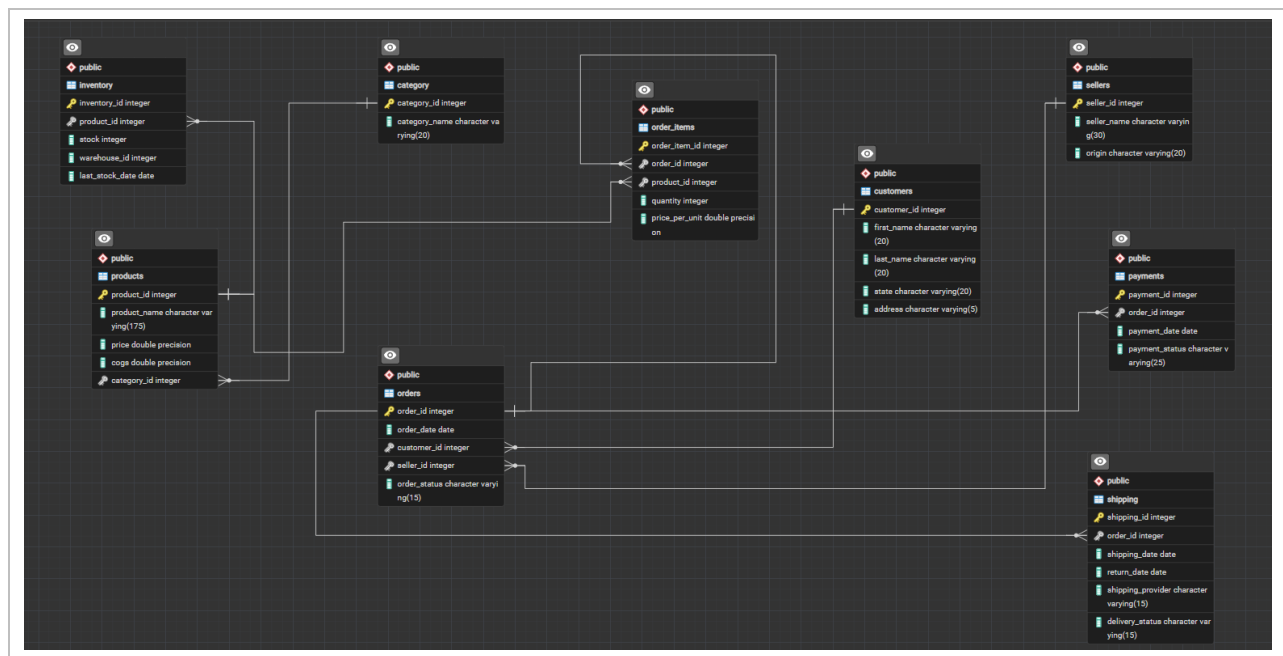
- **order_items:** stores the products, quantity and price per unit of each order.
- **payments:** stores the payment date and payment status of each order.
- **shipping:** stores the shipping date, return date, shipping provider and delivery status of each order.
- **inventory:** stores the stock of each product by warehouse along with the last stock date.

2.4 Table Relationships

The tables are connected with primary key and foreign key relationships. Each relationship is one-to-many, from the parent table to the child table:

Parent Table (PK)	Child Table (FK)	Relationship
category	products	category_id: one category has many products
customers	orders	customer_id: one customer places many orders
sellers	orders	seller_id: one seller fulfills many orders
orders	order_items	order_id: one order has many order items
products	order_items	product_id: one product appears in many order items
orders	payments	order_id: one order is linked to its payment records
orders	shipping	order_id: one order is linked to its shipping records
products	inventory	product_id: one product is stored in inventory records by warehouse

The diagram below shows the SQL database schema with all 9 tables and these relationships.



SQL database schema (amazon_db)

2.5 Data Preparation Notes

Because of the primary key and foreign key relationships, some source data had to be cleaned before it could be imported into the tables, using Python or Excel:

- The products file had invalid values (text and null) in the price and cogs columns, which were cleaned before importing.
- Some product_ids present in the order_items file were not present in the products table, and this was fixed before importing.
- Some product_ids present in the inventory file were not present in the products table, and this was fixed before importing.

2.6 Fact and Dimension Tables (Python)

The SQL database (amazon_db) was used to answer the business questions. For Power BI, a Python script using pandas and SQLAlchemy converts the 9 database tables into a star schema of dimension and fact tables, and writes them

to a second PostgreSQL database (amazon_fact&dim_db). The script performs these steps:

- Reads all 9 tables from amazon_db into pandas dataframes and checks the top 5 rows of each table.
- Builds the dimension tables and the fact tables listed below.
- Fixes the date columns (last_stock_date, order_date, shipping_date, return_date and payment_date) so that they have a proper date data type.
- Writes all 7 resulting tables into the new database.

Table	Type	Built From
dim_products	Dimension	products table joined with the category table on category_id
dim_customers	Dimension	customers table
dim_sellers	Dimension	sellers table
dim_warehouse	Dimension	unique warehouse_id values from the inventory table
dim_date	Dimension	daily calendar from the earliest to the latest date across orders, payments, shipping, returns and inventory, with year, month and day columns
fact_sales	Fact	orders table combined with order_items, payments and shipping using the common order_id column
fact_inventory	Fact	copy of the inventory table

The fact_sales table keeps these columns: order_id, order_date, customer_id, product_id, quantity, price_per_unit, payment_date, payment_status, seller_id, shipping_provider, shipping_date, return_date, order_status and delivery_status. To keep the star schema simple for Power BI, no separate dimension tables were created for orders, order_items, payments and shipping. They are all combined into fact_sales through order_id.

Data Checks in the Script

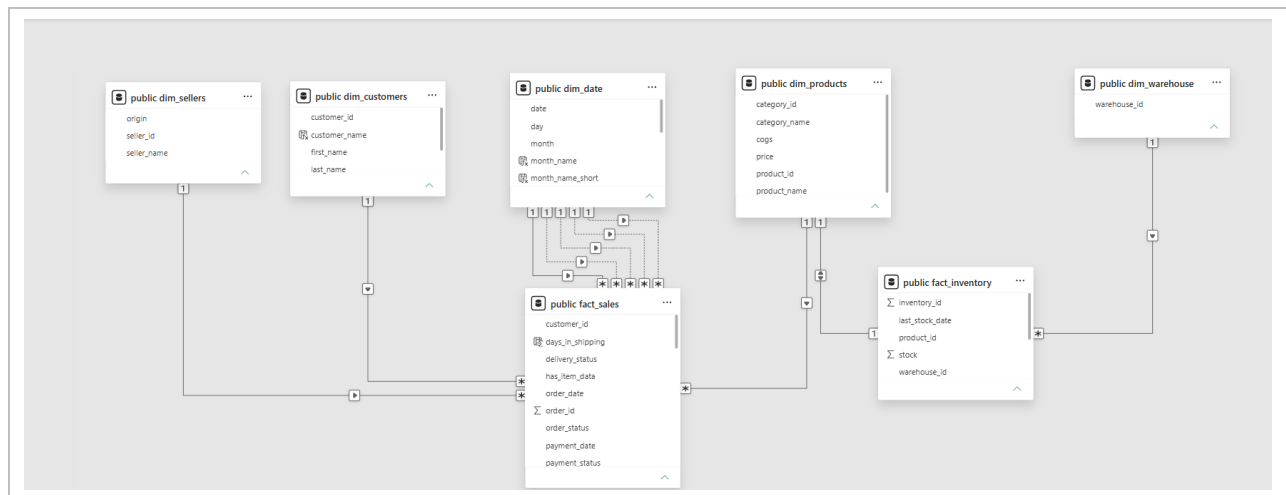
- **Orders without item data:** there are 231 orders present in the orders table that were not recorded in the order_items table. They show null values in product_id, quantity and price_per_unit in fact_sales. A new column, has_item_data (Yes or No), was added to flag these orders.
- **Null return dates:** the script confirmed that null values in return_date are there because those orders were not returned (they were delivered, cancelled or still in transit), and not because of any join error.

2.7 Power BI Data Model

The dimension and fact tables are loaded into Power BI as a star schema. Each dimension table has a one-to-many relationship with the fact tables:

- dim_sellers is connected to fact_sales through seller_id.
- dim_customers is connected to fact_sales through customer_id.
- dim_products is connected to fact_sales and fact_inventory through product_id.
- dim_warehouse is connected to fact_inventory through warehouse_id.
- dim_date is connected to the date columns of the fact tables. One relationship is active and the others are shown as dotted inactive relationships.

Calculated columns such as customer_name, days_in_shipping and the month names were added in Power BI.



Power BI data model (star schema)

3. Business Questions and Insights

This section presents the business questions for which insights were drawn from the SQL analysis, along with the insight for each question.

Question 1

Top 10 selling products based on orders that were not cancelled or returned (include category, product name, total orders, quantity sold and sales value).

Insight: The product "Google Mechanical Keyboard" is the top seller, and all of the top 10 products by sales come from the category "Electronics", indicating high demand for electronic devices.

Question 2

Revenue by each product category.

Insight: The category "Electronics" has the highest sales of \$19,803,860.87 among all the categories, with a contribution of 76.75% to total sales.

Question 3

Average Order Value (AOV) for customers having at least 15 orders.

Insight: There are more than 11 customers who have an average order value of more than 30,000 while having a total number of orders of 15 or more.

Question 5

Find the customers who are registered but have not placed any order yet.

Insight: There are no customers who are registered and have not placed any order.

Question 6

Find the top 3 best-selling product categories in each state, including their sales value.

Insight: The categories "Electronics" and "Home & Kitchen" appear at rank 1 and rank 2 respectively in terms of sales in every state, which describes a constant high demand for these categories across all the states.

Question 7

Find the least-selling product category in each state, including its sales value.

Insight: The category "Pet Supplies" appears as the least-selling product category in most of the states, showcasing its low demand among many states, which is justified because not everyone owns a pet.

Question 9

Find products with stock levels below a certain threshold (for example 10 units) in the inventory, to give a low stock alert.

Insight: Many products that have stock levels below 10 are from Warehouse 1. Warehouse 1 also shows the highest sales volume among all the warehouses, which is confirmed by the warehouse-wise sales check. This shows the low stock levels in Warehouse 1 are likely caused by high product demand, since items there are selling very quickly. To stop stockouts it is necessary to monitor inventory and to replenish stock more often.

Question 10

Identify orders where the shipping delay is more than 5 days since the order was placed, including customer details, order details and shipping provider.

Insight: There are no non-cancelled orders that were delayed in shipping by more than 5 days from the order date. This indicates that, in the recorded dataset, orders were generally shipped within 5 days of being placed.

Question 11

Calculate the percentage of each payment type across all orders, including all the different payment statuses.

Insight: 54.36% of payments came out to be successful, while 35.42% are still pending. This means that a large number of payments have not yet been completed, which could lead to delays in order processing.

Question 13

Calculate the gross profit and profit margin for each product and rank products by their profit margin, while also showing product category, product name, purchase price, selling price, quantity sold and total sales.

Insight: There are more than 176 products which have a profit margin higher than 60%. A planned marketing campaign and promotion strategy to increase their sales can significantly improve the overall profit of the business.

Question 14

Identify the top 10 most returned products and determine what percentage of their ordered units was returned.

Insight: The return rate for all 10 products exceeds 10%, indicating that a significant number of these items are being returned. The business should pay attention to customer reviews and discuss the potential reasons for these returns with the sellers to identify ways to reduce the return rate.

Question 15

Identify sellers who have not made any sales in the last 6 months, and also provide details regarding the date of their most recent sale and their total sales.

Insight: There are no sellers who have been inactive during the last 6 months based on the latest recorded sale date, indicating that all sellers have contributed to sales during this period.

Question 16

Analyse the return activity of customers who placed at least 10 orders in the last 2 years. Calculate their total orders, returned orders and return rate, and classify customers with a return rate of 40% or higher as "High Return Frequency" and others as "Normal Return Frequency".

Insight: There are 19 customers who have a high return frequency. The team should contact them and analyse the reason behind their high return frequency, as it could also affect other future customers in the same residential area.

Question 19

Analyse the performance of each shipping provider by calculating total revenue, the number of orders handled and the average shipping time.

Insight: The average shipping time for each shipping provider is approximately two days. Among all the providers, FedEx handles the highest volume of orders (45.23% of the total) and generates the most sales. This indicates that FedEx manages a significant portion of our orders while maintaining consistent delivery times, resulting in superior sales performance.

Question 20

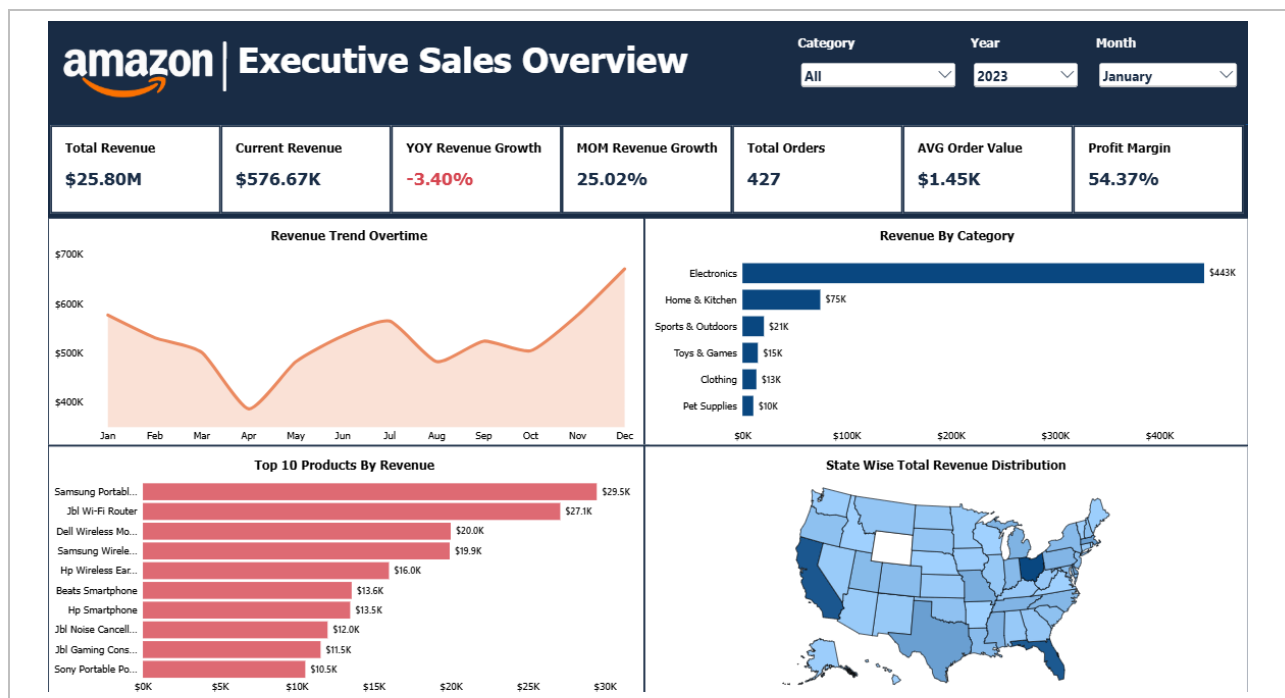
Find the year-over-year revenue growth percentage for each recorded year.

Insight: The business experienced a 3.38% growth in sales in 2021, but faced a 6.77% decline in 2022. In 2023, sales started to recover, growing by 1.14%, although the growth was relatively small. The team should investigate the reasons behind the decline in 2022 and develop strategies to address those issues so that a similar decline can be avoided in the coming years.

4. Power BI Dashboard

The Power BI dashboard is built on the star schema data model described in Section 2.7 and has four pages. Each page has Category, Year and Month slicers at the top and a row of KPI cards, followed by charts and tables. The screenshots below show the pages with Category set to “All”, Year set to “2023” and Month set to “January”.

4.1 Executive Sales Overview



Page 1: Executive Sales Overview

- **KPIs:** total revenue is \$25.80M, with current revenue of \$576.67K. Year-over-year revenue growth is -3.40% and month-over-month growth is 25.02%. There are 427 orders, the average order value is \$1.45K and the profit margin is 54.37%.
- **Revenue trend:** revenue starts the year near \$580K, falls to its lowest point of about \$380K in April, then recovers and rises to its highest point of about \$670K in December.

- **Revenue by category:** Electronics leads with \$443K, far ahead of Home & Kitchen (\$75K), Sports & Outdoors (\$21K), Toys & Games (\$15K), Clothing (\$13K) and Pet Supplies (\$10K).
- **Top 10 products:** the top products by revenue are all electronic items, led by the Samsung Portable product (\$29.5K), the JBL Wi-Fi Router (\$27.1K), the Dell Wireless Mouse (\$20.0K) and the Samsung Wireless product (\$19.9K).
- **State-wise revenue:** California, Ohio and Florida are shaded darkest on the map, showing the highest revenue among the states.

4.2 Customers & Sellers Insights



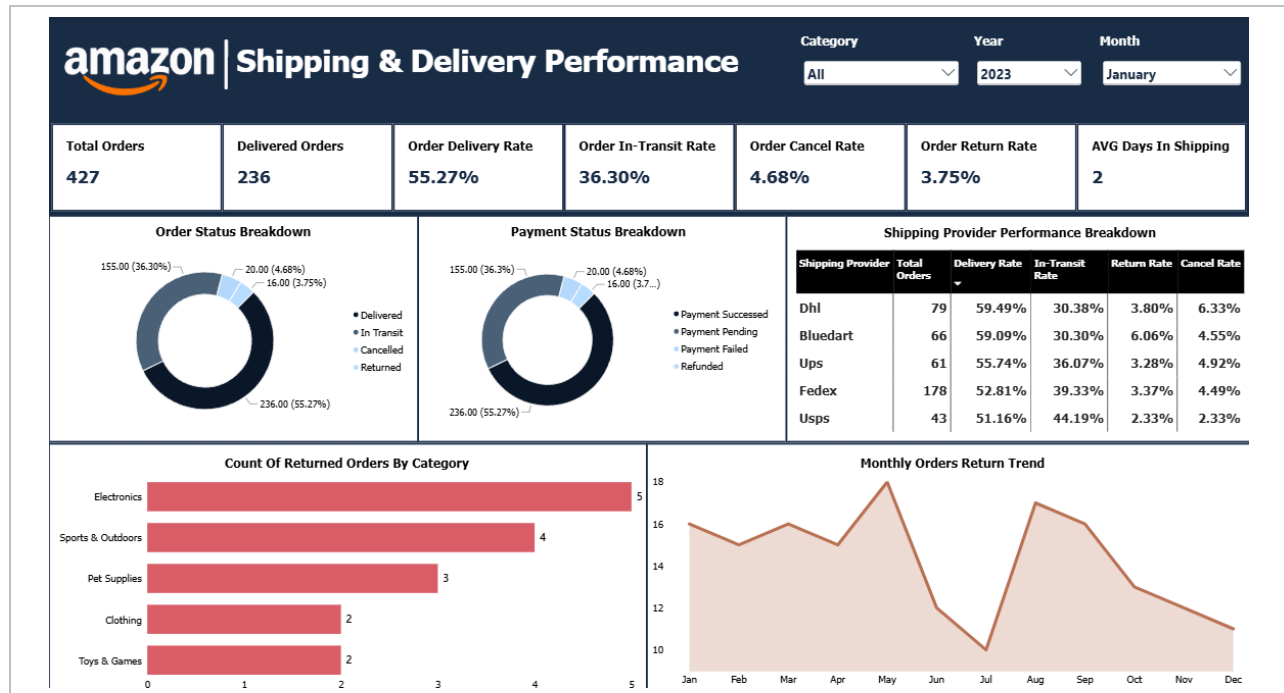
Page 2: Customers & Sellers Insights

- **KPIs:** there are 900 customers, of which 331 are returning customers, and 54 active sellers. Average revenue per seller is \$10.68K and average revenue per customer is \$640.75.
- **Customers by state:** California has the most customers (71), followed by Texas (57), Florida (55) and Ohio (45). New York and Pennsylvania have

34 each and Illinois has 30. Georgia, Massachusetts, North Carolina and Washington have 25 each.

- **Top customers by revenue:** Janet Thomas is first with \$22,583.05 (3.92% of revenue), followed by Megan Diaz with \$21,968.28 (3.81%), Jacqueline Davis with \$21,050.95 (3.65%), Dennis Chavez with \$19,922.20 (3.45%) and Jordan Brown with \$13,698.86 (2.38%). Most of these customers generate their revenue from only one to three orders.
- **Return frequency:** customers such as Deborah Ramirez, Dennis Sanders, Jerry Gray, John Mitchell and Madison Hernandez are segmented as "High Frequency" returners, with return rates between 50% and 100%.
- **Seller origin:** the 54 sellers come from many countries. The USA has the largest share, followed by China and Japan, and then Brazil, Canada, Germany, India and others.
- **Seller performance:** Roborock earns the highest revenue (\$50,279.91), followed by Yeti Coolers (\$30,509.11), Amazonbasics (\$29,582.63), Corkcicle (\$25,513.18) and Wyze Labs (\$24,865.62). Roborock reaches the top with only 10 orders.

4.3 Shipping & Delivery Performance

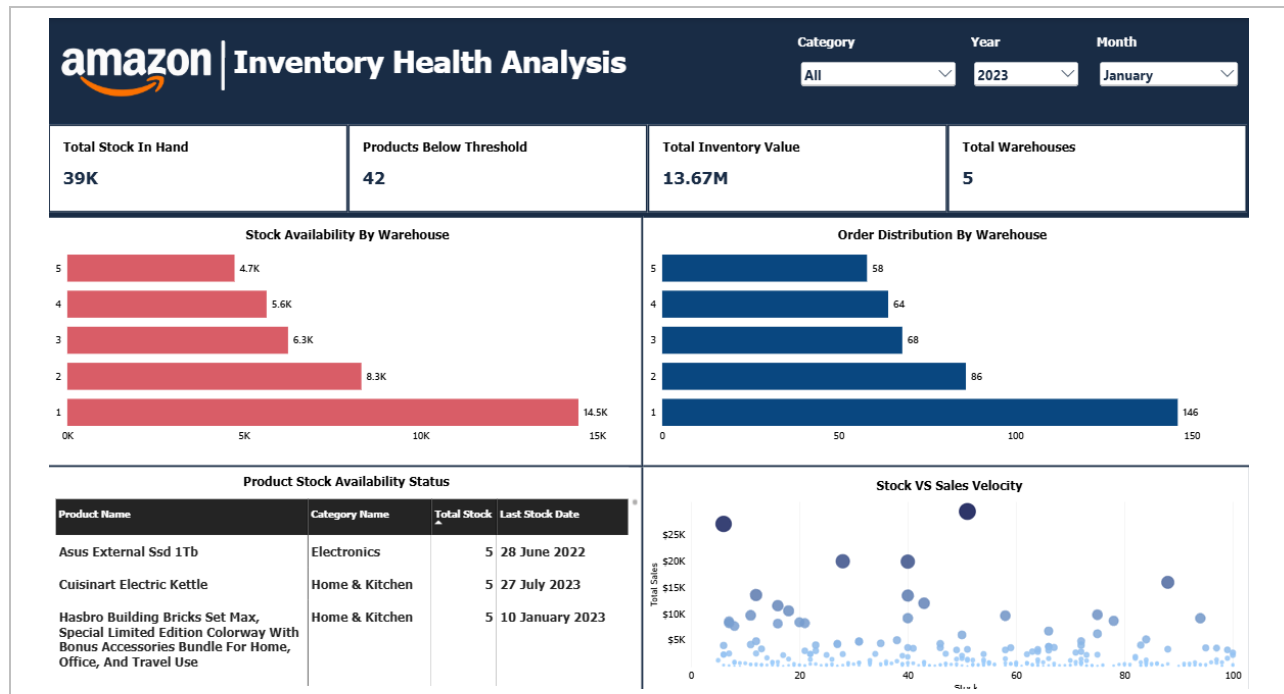


Page 3: Shipping & Delivery Performance

- **KPIs:** of 427 orders, 236 are delivered. The delivery rate is 55.27%, the in-transit rate is 36.30%, the cancel rate is 4.68% and the return rate is 3.75%. The average time in shipping is 2 days.
- **Order status:** Delivered 236 (55.27%), In Transit 155 (36.30%), Cancelled 20 (4.68%) and Returned 16 (3.75%).
- **Payment status:** Payment Successful 236 (55.27%), Payment Pending 155 (36.30%), Payment Failed 20 (4.68%) and Refunded 16 (3.75%).
- **Shipping providers:** FedEx handles the most orders (178), followed by DHL (79), Bluedart (66), UPS (61) and USPS (43). DHL has the highest delivery rate (59.49%) and USPS has the lowest (51.16%) with the highest in-transit rate (44.19%). Bluedart has the highest return rate (6.06%) and DHL the highest cancel rate (6.33%).
- **Returns by category:** Electronics has the most returned orders (5), followed by Sports & Outdoors (4), Pet Supplies (3), and Clothing and Toys & Games (2 each).

- **Monthly return trend:** returns peak in May (about 18), fall to the lowest point in July (about 10), rise again in August (about 17) and then decline towards December (about 11).

4.4 Inventory Health Analysis



Page 4: Inventory Health Analysis

- **KPIs:** total stock in hand is 39K units across 5 warehouses, with a total inventory value of \$13.67M. 42 products are below the stock threshold.
- **Stock by warehouse:** Warehouse 1 holds the most stock (14.5K), followed by Warehouse 2 (8.3K), Warehouse 3 (6.3K), Warehouse 4 (5.6K) and Warehouse 5 (4.7K).
- **Orders by warehouse:** Warehouse 1 also handles the most orders (146), followed by Warehouse 2 (86), Warehouse 3 (68), Warehouse 4 (64) and Warehouse 5 (58). Warehouse 1 carries the highest demand, so it needs close stock monitoring and frequent replenishment.
- **Product stock status:** the table lists products with very low stock, for example the Asus External SSD 1TB (Electronics), the Cuisinart Electric

Kettle (Home & Kitchen) and a Hasbro Building Bricks Set (Home & Kitchen), each with only 5 units in stock.

- **Stock vs sales velocity:** most products sit at the bottom of the chart with low sales. A few products reach \$20K to \$27K in sales while holding only a low to moderate stock, so these fast-selling products are the first candidates for restocking.